



River Trothy nutrient trends

A report on behalf of the Wye Salmon Association.

Version	1.0
Date	24-April-26
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A brief analysis of NRW water quality data from the Trothy sub catchment, examining trend in TON & RP between 2000-2025.

Introduction

A recent report on the Trothy sub catchment (Green, 2026) highlighted a substantial conversion of acid grassland to improved grassland occurring between 2000 and 2023. This begs the question of whether this apparent intensification of agricultural activity led to an increase in nutrient levels in the river system. This report reviews EA/NRW water quality data downloaded from the NRW website, covering the period 2000 – 2025. Specifically, we shall examine the long-term trends for total oxidised nitrogen (TON) and orthophosphate (RP).

Results and discussion

Reasonably complete datasets are available for the Llanymynach and Llymon Brooks, plus the R. Trothy at several locations from Caggle Street in the upper river, to Troy House close to the Wye confluence. As always with NRW and EA monitoring, there is a switch from monthly sampling before 2010 to 3 or 4 times a year, or not at all, after this date. In addition, there is a hole in the data between circa 2014 and circa 2019, the reason for which is not known to the author.

We shall begin our analysis on the lower Trothy, then work upstream, ending with the two tributaries.

Trothy at Troy House

Troy House is at the bottom of the sub catchment, just above the Wye confluence. Figure 1 shows the annual average TON trend at this location. This shows no significant change over the period of interest, and the long-term average is consistent with the CS test strip data.

Figure 2 shows the annual average RP trend. This also shows no change since 2000 and, similarly, the long-term average is a good match to CS Hanna data.

I have not included the charts, but a very similar picture emerges at Dingestow, a few km upstream. Both TON and RP are essentially unchanged between 2000 and 2025, with average levels replicating our CS data.

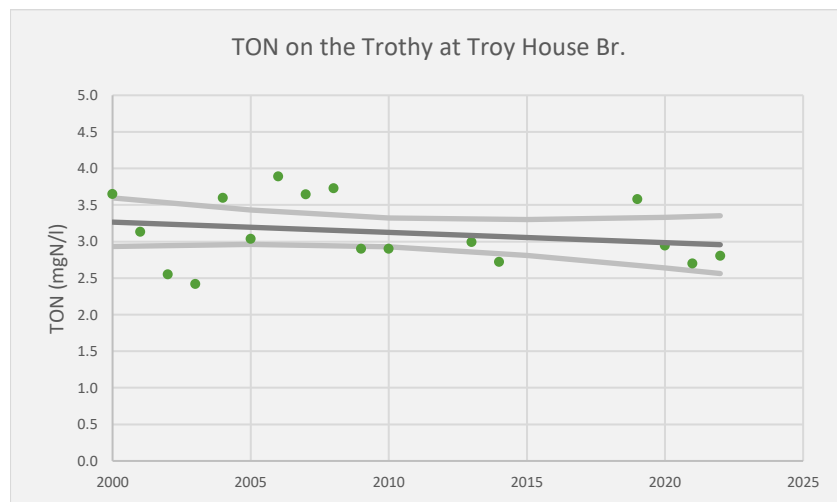


Figure 1: annual average TON on the Trothy at Troy House

Dark grey line is a linear regression; light grey shows 90% confidence limits.

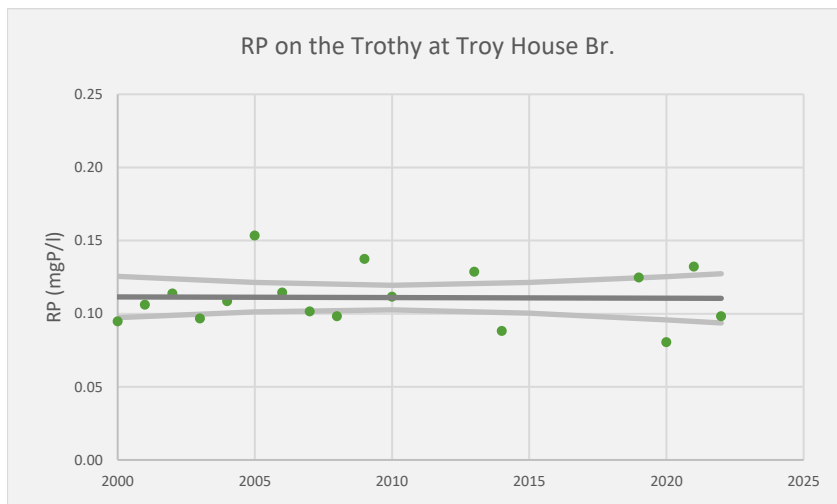


Figure 2: annual average RP on the Trothy at Troy House.

Dark grey line is a linear regression; light grey shows 90% confidence limits.

Trothy at Onen

Onen is in the middle river, above the Llymon confluence. Figures 3 & 4 show the trend plots.

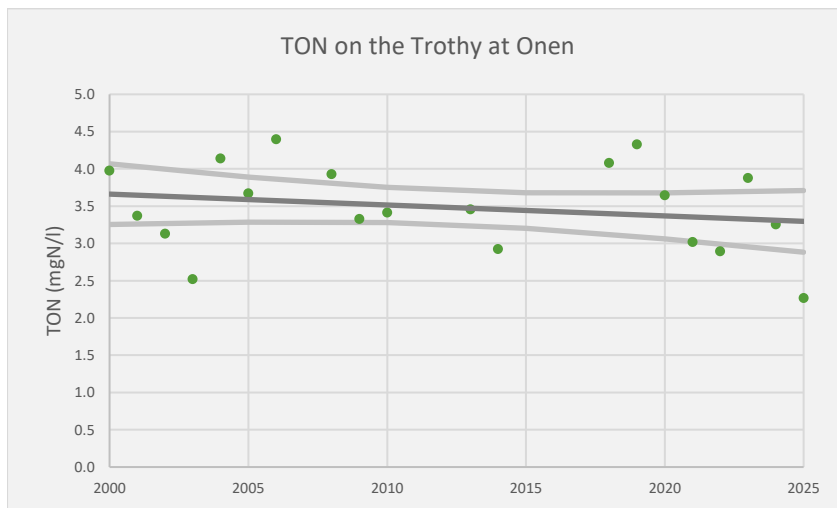


Figure 3: annual average TON on the Trothy at Onen.

Dark grey line is a linear regression; light grey shows 90% confidence limits.

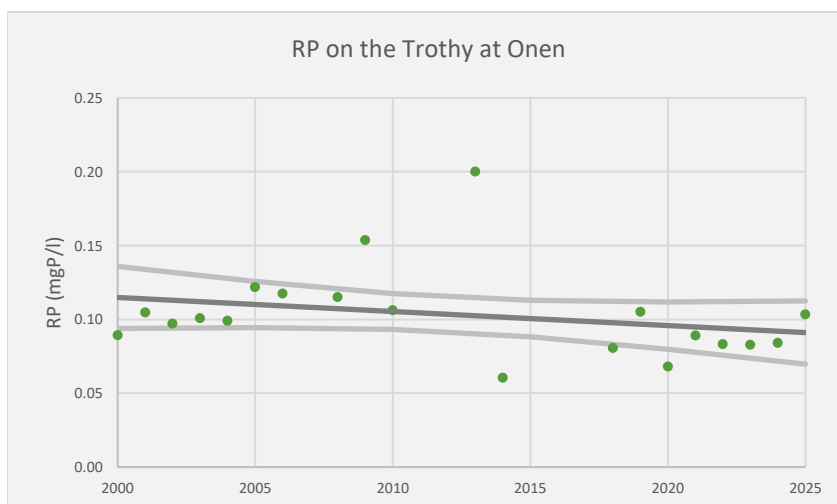


Figure 4: annual average RP on the Trothy at Onen.

Dark grey line is a linear regression; light grey shows 90% confidence limits.

We find a slightly higher TON value up at Onen but, once more no significant long-term change. RP is marginally lower than in the lower river but again shows little long-term change. Note that the low readings of 2013 and 2020 were each based on only 2 measurements: hardly statistically robust.

Trothy at Caggle Street Bridge

Caggle Street is in the upper catchment, 5km or so above the Llanymynach confluence.

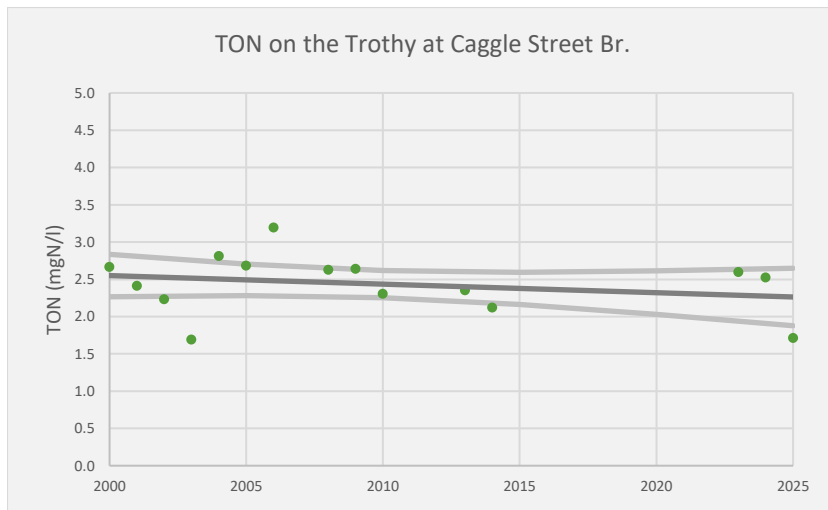


Figure 5: annual average TON on the Trothy at Caggle Street.

Dark grey line is a linear regression; light grey shows 90% confidence limits.

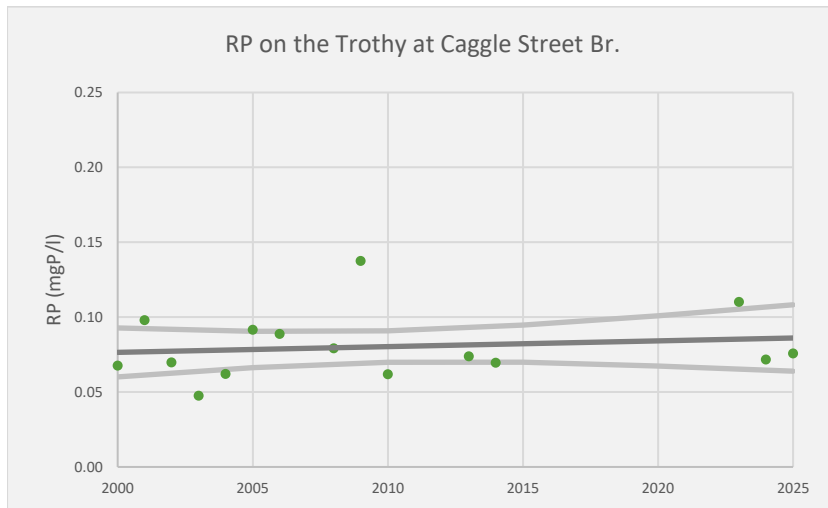


Figure 6: annual average RP on the Trothy at Caggle Street.

Dark grey line is a linear regression; light grey shows 90% confidence limits.

Figure 5 shows average TON values to be lower here (above the Llanymynach) but the trend is steady. Likewise, figure 6 shows RP values to be lower than further downstream but again exhibiting no long-term change.

Llanymynach Brook

The corner of the sub catchment which is drained by the Llanymynach has a notable concentration of arable land (see the map in Green, 2026). Thus, we might expect something different. In figure 7, we indeed find a substantially higher TON value, very similar to those found on the Worm for example. But the long-term trend is steady.

The RP chart in figure 8 is decidedly weird, showing a substantial decline. Current values are in line with the rest of the sub catchment but, back in the early 2000s, RP levels were twice what they are today. The monthly data (not shown here) for 2004 & 2005 exhibits none of the clear seasonality which one might expect from a fixed flow source. Also, the fact that the high RP took a couple of decades to clean up is perhaps more suggestive of a diffuse origin. Whatever occurred, current RP levels on this tributary appear to be in line with the rest of the upper sub catchment (e.g. Caggle Street above).

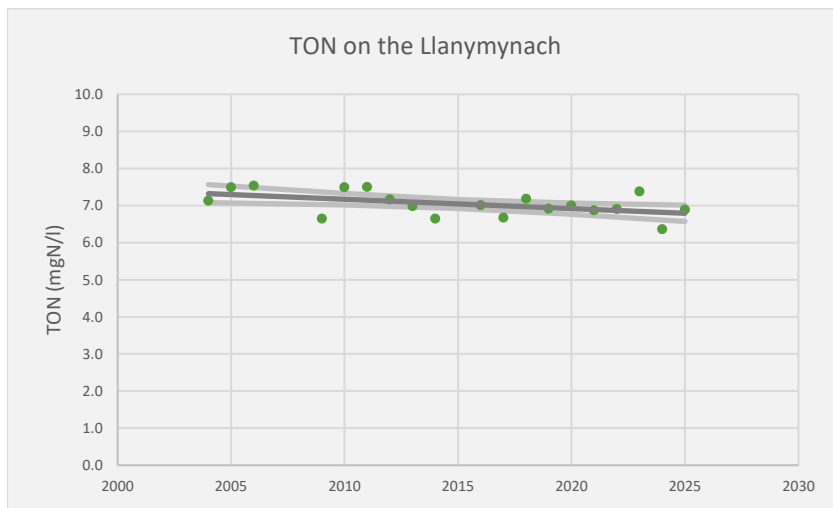


Figure 7: annual average TON on the Llanymynach Brook.

Note the change of scale on the vertical axis of this plot.

Dark grey line is a linear regression; light grey shows 90% confidence limits.

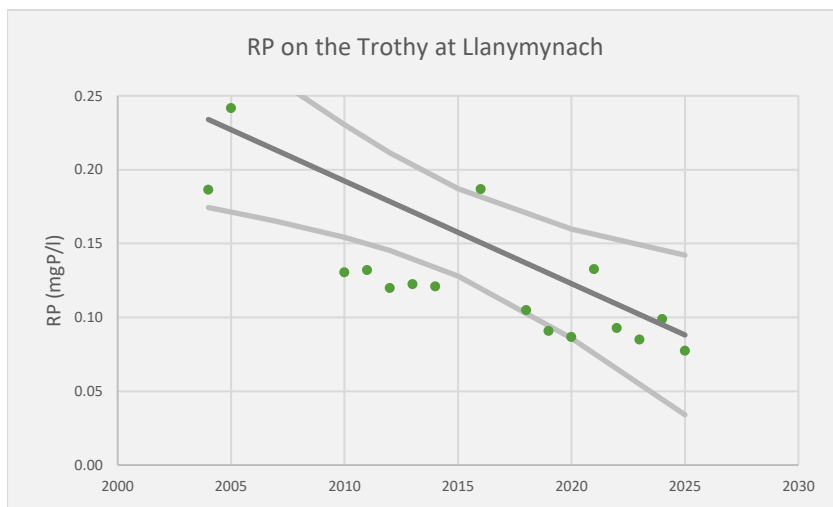


Figure 8: annual average RP on the Llanymynach Brook.

Dark grey line is a linear regression; light grey shows 90% confidence limits.

Llymon Brook

In the Llymon catchment, we are back to the mostly-pasture scenario, typical of most of the Trothy sub catchment. TON levels (figure 9) are modest and steady. RP levels (figure 10) are on the low side, being similar to Onen say, and steady.

Looking at the Llymon plots, it is interesting in passing to pick out the low outlier in 2003, which occurs on both parameters. Going back to the other charts, we can pick out similar dips in 2002-3 on several. This period was immediately after the 2001 foot and mouth epidemic which saw a massive reduction in the amount of manure going onto pasture. Which had an immediate and highly significant impact on river nutrient levels in many parts of the Wye catchment.

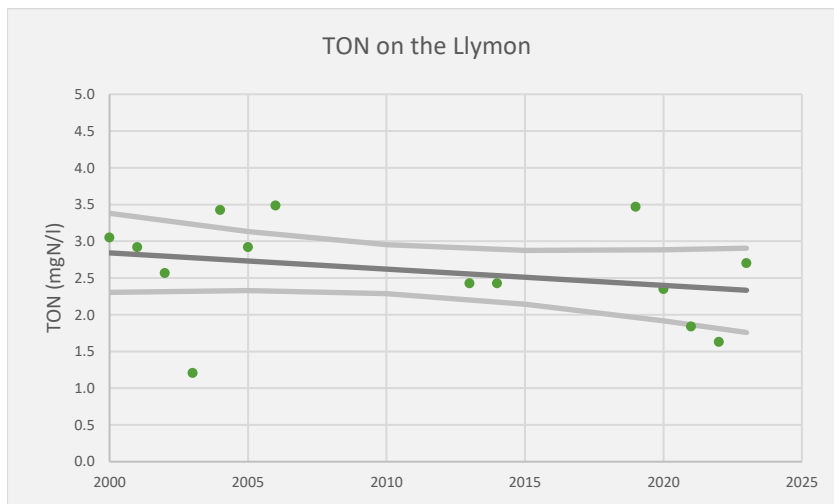


Figure 9: annual average TON on the Llymon Brook.

Dark grey line is a linear regression; light grey shows 90% confidence limits.

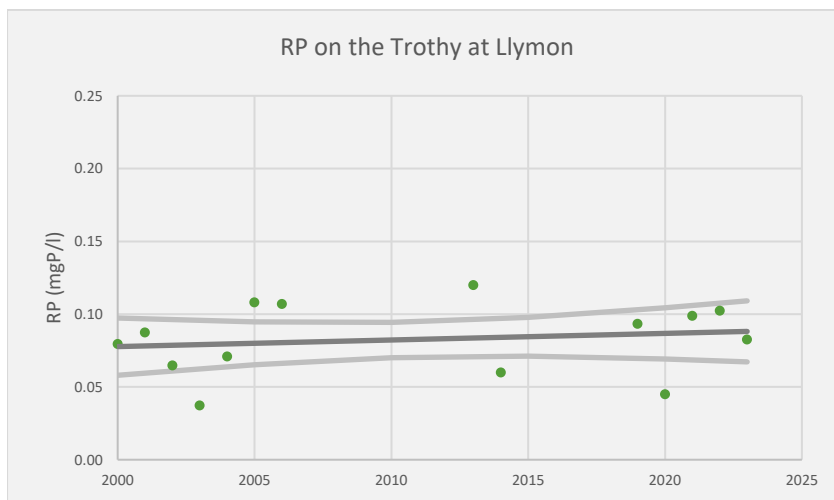


Figure 10: annual average RP on the Llymon Brook.

Dark grey line is a linear regression; light grey shows 90% confidence limits.

Conclusion

- There has been no statistically significant change in TON or RP concentrations within the Trothy sub catchment over the last 25 years. In an area largely devoid of IPUs, nutrient loads have remained remarkably constant. Thus, the change in land classification from acid grassland to improved grassland does not appear to be associated with a significant increase in livestock numbers. It may simply be that reclassification was mostly about maximising CAP grants, for example.
- The statistically significant drop in TON and RP concentrations in 2002-3 does show what happens when livestock farmers put less manure on their land. The rivers clean up rapidly.
- The highest location for TON (average circa 7 mgN/l) seen on the Llanymynach correlates with a notable concentration of arable farming. Other areas, being pasture dominated, are generally in the 2.5-3.5 mgN/l range.
- The Llanymynach had very high levels of RP in the early 2000s which has since cleaned up to typical Trothy levels (circa 0.1 mgP/l).
- Annual average TON and RP concentrations are very consistent with our CS data.

References

Green, G. (2026). *Trothy 23-25*.